

Substrate-Driven Localization Gaps in Pipe Crack Detection

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Abstract—Automated visual inspection of piping systems is critical for structural integrity monitoring, yet the influence of substrate visual properties on detection performance remains poorly understood. This paper investigates how substrate complexity affects crack detection difficulty when training data quantity is controlled. Three crack specimen categories were fabricated on substrates spanning a texture gradient—smooth PVC to porous tissue—and a class-balanced dataset (400 labels per class) was created. YOLO-World XL and YOLOv8x were fine-tuned across five random seeds on an NVIDIA DGX A100. YOLO-World XL achieved 0.863 mAP@50-95 (standard deviation 0.004), yet per-class AP ranged from 94.2% (dummy crack on smooth PVC) to 73.9% (paper crack on textured tissue)—a 20.2 percentage point gap consistent across both architectures. Decomposing by IoU threshold reveals this is primarily a localization problem: paper crack maintains 97.1% AP@50 but drops 23.1 percentage points at stricter thresholds. Three augmentation-based mitigation strategies (CutMix, multi-scale training, and their combination) failed to significantly reduce the gap (Bonferroni-corrected p -values above 0.05), indicating the disparity reflects a fundamental localization challenge tied to substrate visual properties rather than insufficient training-time diversity.

Index Terms—YOLO-World, object detection, crack detection, pipe inspection, transfer learning, fine-tuning, deep learning, substrate analysis, data augmentation

I. INTRODUCTION

Structural integrity monitoring of piping systems is a critical safety concern across sectors including oil and gas, water treatment, and manufacturing. Traditional manual inspection methods are labor-intensive, subjective, and often limited to accessible areas. Automated visual inspection based on deep learning offers a promising alternative, yet real-world piping systems comprise diverse materials and surface conditions. Without quantifying how substrate properties affect detection, practitioners risk deploying models with uncharacterized failure modes on specific pipe materials.

YOLO-World [1] extends the YOLO [2] single-stage detection framework with open-vocabulary capabilities through vision-language pre-training, enabling strong transfer to custom categories with limited labeled data. Its pre-trained representations should, in principle, generalize across surface types, making it suitable for investigating whether substrate-driven performance differences persist even when training data is controlled.

However, existing studies report aggregate metrics across a single surface type, implicitly assuming that balanced data yields comparable per-class performance [3]–[6]. While it is intuitive that textured backgrounds degrade detection, the *magnitude* of substrate-dependent variation under controlled, class-balanced conditions—and whether standard augmentation can mitigate it—remain unknown. Ma et al. [7] showed that background complexity degrades performance, but evaluated a single surface without controlling for class balance.

This paper addresses: **How does the visual complexity of the defect substrate affect detection performance when training data quantity is controlled?** The contributions are:

- 1) Quantitative evidence that defect-substrate visual properties produce a 20–22 percentage point (pp) per-class Average Precision (AP) gap—consistent across YOLO-World XL and YOLOv8x over five seeds and not reducible to data imbalance.
- 2) An Intersection-over-Union (IoU) threshold decomposition revealing a *localization* problem: paper crack retains 97.1% AP@50 but drops 23.1 pp at stricter thresholds, whereas dummy crack drops only 3.9 pp.
- 3) Evidence that three augmentation strategies fail to close the gap (Bonferroni-corrected $p > 0.05$), indicating a challenge beyond training-time diversity.

II. RELATED WORK

A. Crack Detection Methods

Crack detection has been extensively studied in civil infrastructure, with Hsieh and Tsai [3] providing a comprehensive review of machine learning approaches spanning classical feature engineering to modern deep learning. Convolutional neural network (CNN)-based methods have largely superseded classical edge detection approaches [4], with Cha et al. [5] achieving 98% accuracy on concrete surfaces using a sliding-window CNN classifier. More recently, Lv et al. [6] deployed lightweight YOLOv8n on amphibious sewer robots, demonstrating real-time inference on embedded hardware. Ma et al. [7] proposed a texture exploration network that explicitly models surface complexity, improving defect detection on textured backgrounds. However, these studies evaluate performance in

aggregate across a single surface type, without systematically examining how substrate properties affect per-class detection difficulty when data quantity is controlled.

B. YOLO-Based Object Detection

The YOLO architecture [2] frames detection as single-stage regression, prioritizing inference speed. YOLOv8 [8] introduced a decoupled head, anchor-free predictions, and an improved feature pyramid network, achieving state-of-the-art accuracy–speed trade-offs. YOLO-World [1] extends the YOLOv8 backbone with a Re-parameterizable Vision-Language Path Aggregation Network (RepVL-PAN) that aligns visual features with CLIP [9] text embeddings, enabling open-vocabulary detection. This vision-language fusion allows the model to leverage semantic priors from natural language, which is particularly relevant for domain-specific tasks where category descriptions (e.g., “crack on PVC pipe”) may guide feature extraction.

C. Data Augmentation for Hard Examples

CutMix [10] generates composite training images by replacing rectangular patches with regions from other samples, encouraging attention to discriminative local features. Multi-scale training varies input resolution across iterations, improving robustness to size variation. Copy-paste augmentation [11] offers stronger instance-level diversity but requires segmentation masks, making it inapplicable to bounding-box-only datasets. We evaluate CutMix and multi-scale training as practical, mask-free mitigation strategies.

D. Transfer Learning for Industrial Inspection

Pre-trained models on COCO [12] and Objects365 [13] provide robust transferable features. However, while class imbalance effects [14], [15] have been studied extensively, per-class performance variation driven by visual property differences—rather than data quantity—remains underexplored.

III. METHODOLOGY

A. Dataset Preparation

Three crack specimen categories were fabricated on blue-colored substrates of varying surface texture. Controlled proxies were chosen deliberately to isolate substrate texture as the independent variable; field specimens would introduce confounding factors (crack age, lighting variation, pipe curvature, and surface contamination) that would prevent attributing performance differences to any single factor. The three categories span a visual complexity gradient from low to high:

- *Dummy cracks*—drawn lines on smooth PVC pipes, providing sharp, high-contrast edges with no surface texture interference.
- *PVC pipe cracks*—physical fractures on smooth PVC, introducing realistic crack morphology while maintaining a uniform substrate.
- *Paper cracks*—cracks on blue-painted tissue rolls, where the porous, fibrous substrate creates visual noise that obscures crack boundaries.

This gradient allows systematic investigation of how increasing substrate complexity affects detection and localization. Fig. 1 shows annotated samples from each category.

The dataset is class-balanced at 400 labels per class (1,200 total), annotated using Roboflow [16]. Preprocessing included auto-orientation, resizing to 640×640 , and contrast adjustment. Augmentation ($3\times$) comprised flips, rotation ($\pm 15^\circ$), shear ($\pm 10^\circ$), brightness ($\pm 25\%$), exposure ($\pm 15\%$), blur (up to 1.5 px), and noise (up to 1%). The augmented dataset (2,617 images) was split into training (88%), validation (8%), and test (4%). Table I summarizes the composition.

TABLE I
DATASET COMPOSITION AND SPLIT CONFIGURATION

Property	Value
Labels per class (Dummy / Paper / PVC)	400 / 400 / 400
Total annotations	1,200
Post-augmentation images	2,617
Training / Validation / Test	2,292 / 217 / 108

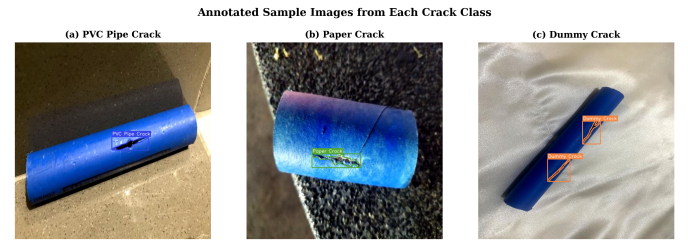


Fig. 1. Annotated samples. (a) PVC pipe crack. (b) Paper crack. (c) Dummy crack.

B. Model Architecture and Training

YOLO-World XL (YOLOv8x-worldv2, 72.9M parameters, 273.3 giga floating-point operations (GFLOPs)) was selected as the primary detector. Its vision-language pre-training via a RepVL-PAN fusing visual features with CLIP [9] text embeddings provides the richest available feature representations in the YOLO family, ensuring that any observed per-class differences reflect substrate complexity rather than model capacity limitations. The model was pre-trained on COCO [12], Objects365 [13], and GoldG (a grounding dataset used in the original YOLO-World training pipeline [1]). A standard YOLOv8x [8] baseline (without open-vocabulary capability) was included to test whether the substrate effect is architecture-specific.

Fine-tuning was performed on an NVIDIA DGX A100-40GB for 100 epochs using AdamW [17] with learning rate 2×10^{-4} , weight decay 0.05, batch sizes of 8 (YOLO-World XL) and 16 (YOLOv8x)—with Ultralytics’ nominal batch size ($nbs = 64$) auto-scaling the effective learning rate—and 640×640 input resolution. All classes were trained jointly under identical conditions with no class-specific hyperparameters, ensuring that per-class differences are attributable to substrate

properties rather than training procedure variations. Early stopping patience was 20 epochs; the best model was selected by validation mAP@50-95. Both models were trained five times each with seeds 0, 1, 2, 42, and 123; dataset splits were held fixed across all 10 runs. The low baseline standard deviations ($\text{std} \leq 0.014$ across all metrics and classes) confirm stable optimization; Fig. 2 shows convergence with no overfitting. Experiment tracking used Weights & Biases [18].

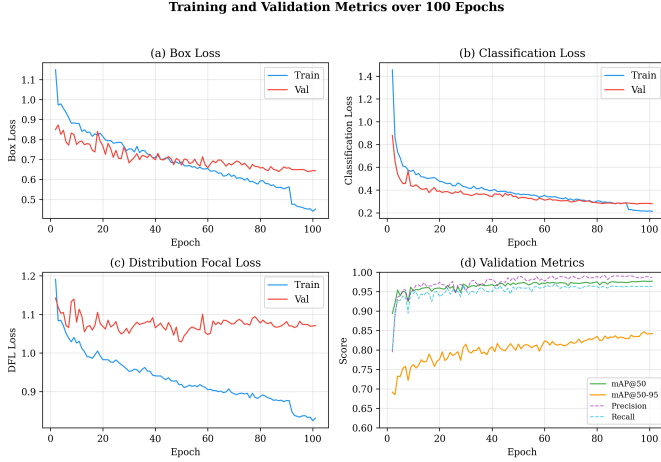


Fig. 2. Training and validation metrics over 100 epochs (YOLO-World XL, seed 42, representative of all five seeds). All losses converge smoothly with no overfitting.

C. Evaluation and Mitigation Strategies

Detection performance is evaluated following the COCO protocol [12] using precision, recall, F1 score, and mAP. The primary metric is mean Average Precision (mAP)@50-95, which averages AP across ten IoU thresholds from 0.50 to 0.95, penalizing imprecise localization more heavily than mAP@50.

To test whether standard augmentation can reduce the substrate-dependent AP gap, three strategies were applied to YOLO-World XL only, keeping all other hyperparameters identical: (1) **CutMix** [10] with probability 0.5; (2) **Multi-scale training** at 960×960 , evaluated at 640×640 ; and (3) **Combined**. Each was trained with the same five seeds for paired comparison. Batch size was reduced to 4 for multi-scale and combined strategies to prevent out-of-memory errors at the higher training resolution.

IV. EXPERIMENTAL RESULTS

A. Overall and Per-Class Performance

For the best single run (seed 42), YOLO-World XL achieves 86.9% mAP@50-95 with precision of 98.4%, recall of 97.7%, and F1 score of 98.1%, indicating strong overall detection capability. However, per-class AP varies substantially: dummy crack at 95.6%, PVC pipe crack at 91.2%, and paper crack at 74.0%—a 21.6 pp gap despite 400 labels per class. This hierarchy aligns directly with the substrate complexity gradient described in Section III-A: smooth, high-contrast surfaces yield

the highest AP, while textured, low-contrast surfaces produce the lowest.

Fig. 3 shows representative detection outputs across all three classes. Bounding boxes on dummy and PVC pipe cracks are tight and well-aligned with the crack boundaries, whereas paper crack predictions exhibit visibly looser localization—particularly at crack endpoints where edges gradually fade into the textured substrate. This visual evidence is consistent with the IoU-threshold decomposition analyzed in Section V-A. Multi-seed averages in Section IV-B confirm a consistent 20–22 pp gap across both architectures, ruling out seed-specific artifacts.



Fig. 3. Detection results from YOLO-World XL (seed 42). Paper crack predictions show looser localization where crack edges blend into the textured substrate.

B. Model Comparison: YOLO-World XL vs. YOLOv8x

Both YOLO-World XL ($\sim 72.9\text{M}$ parameters) and YOLOv8x ($\sim 68.2\text{M}$ parameters) were compared across five seeds (Table II, Fig. 4). YOLO-World XL achieves higher mean values across all metrics, with statistically significant improvements in mAP@50 ($p = 0.038$), recall ($p = 0.024$), and F1 ($p = 0.025$); mAP@50-95 and precision show large effect sizes (Cohen’s d 1.06–1.35) but do not reach significance at $n = 5$, reflecting limited statistical power rather than negligible differences.

Critically, the per-class performance hierarchy is identical across both architectures: dummy crack achieves the highest AP, followed by PVC pipe crack, with paper crack consistently the lowest. The Dummy–Paper AP gap is 20.2 pp for YOLO-World XL and 21.5 pp for YOLOv8x, confirming the substrate effect is architecture-independent. This consistency across models with different pre-training strategies (vision-language vs. vision-only) strengthens the conclusion that the performance gap is driven by substrate properties rather than model-specific limitations. Both models exceed real-time rates (>37 frames per second (FPS) on A100 at 640×640 input, batch size 1, measured via Ultralytics’ built-in profiler), making them suitable for deployment in automated inspection systems.

TABLE II
YOLO-WORLD XL VS. YOLOV8X (MEAN $\pm$ STD, 5 SEEDS)

Metric	YOLO-World XL	YOLOv8x	p	d
mAP@50-95	0.8628 $\pm$ 0.0044	0.8545 $\pm$ 0.0075	0.124	1.35
mAP@50	0.9862 $\pm$ 0.0019	0.9807 $\pm$ 0.0056	0.038*	1.31
Precision	0.9849 $\pm$ 0.0040	0.9787 $\pm$ 0.0072	0.157	1.06
Recall	0.9803 $\pm$ 0.0047	0.9686 $\pm$ 0.0109	0.024*	1.40
F1 Score	0.9826 $\pm$ 0.0040	0.9736 $\pm$ 0.0079	0.025*	1.44
<i>Per-class AP@50-95</i>				
Dummy crack	0.9415 $\pm$ 0.0096	0.9396 $\pm$ 0.0084	—	—
PVC pipe crack	0.9077 $\pm$ 0.0047	0.8996 $\pm$ 0.0140	—	—
Paper crack	0.7393 $\pm$ 0.0067	0.7243 $\pm$ 0.0134	—	—

* $p < 0.05$ (paired t -test, $n = 5$ seeds). d : Cohen's d (pooled SD).
Per-class statistics omitted due to insufficient power at $n = 5$.

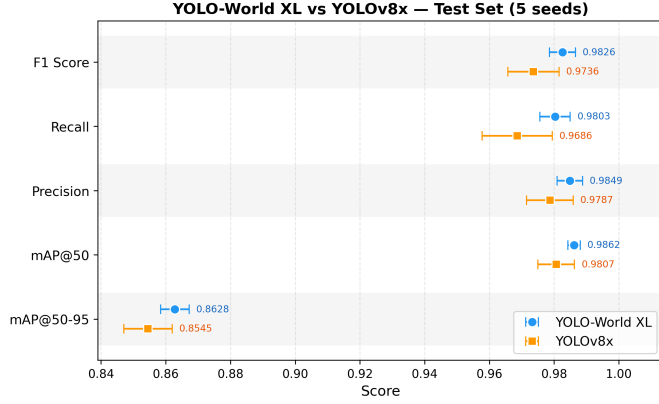


Fig. 4. YOLO-World XL vs. YOLOv8x (mean $\pm$ std, 5 seeds). YOLO-World XL achieves higher means with lower variance; three of five differences reach $p < 0.05$.

C. Mitigation Strategy Evaluation

None of the three strategies improved performance (Table III, Figs. 5–6): paired t -tests ($n = 5$ seeds) showed CutMix reduced mAP@50-95 by 2.3 pp ($p = 0.085$), Multi-Scale by 2.4 pp ($p = 0.147$), and Combined by 5.8 pp ($p = 0.115$), all non-significant after Bonferroni correction ($\alpha \approx 0.0167$). The substrate gap was not reduced: Multi-Scale achieved the largest change (20.2 $\rightarrow$ 18.1 pp, $p = 0.334$), while Combined reduced it to 18.4 pp at the cost of degraded overall performance.

TABLE III
MITIGATION STRATEGY EVALUATION (MEAN $\pm$ STD, 5 SEEDS).
YOLO-WORLD XL ONLY.

Metric	Baseline	CutMix	Multi-Scale	Combined
mAP@50-95	0.8628 $\pm$ 0.0044	0.8399 $\pm$ 0.0218	0.8390 $\pm$ 0.0284	0.8045 $\pm$ 0.0657
mAP@50	0.9862 $\pm$ 0.0019	0.9852 $\pm$ 0.0025	0.9786 $\pm$ 0.0081	0.9570 $\pm$ 0.0506
Precision	0.9849 $\pm$ 0.0040	0.9792 $\pm$ 0.0081	0.9718 $\pm$ 0.0111	0.9506 $\pm$ 0.0458
Recall	0.9803 $\pm$ 0.0047	0.9759 $\pm$ 0.0048	0.9668 $\pm$ 0.0120	0.9417 $\pm$ 0.0606
F1 Score	0.9826 $\pm$ 0.0040	0.9776 $\pm$ 0.0060	0.9693 $\pm$ 0.0108	0.9461 $\pm$ 0.0532
<i>Per-class AP@50-95</i>				
Dummy crack	0.9415 $\pm$ 0.0096	0.9241 $\pm$ 0.0123	0.9048 $\pm$ 0.0488	0.8816 $\pm$ 0.0488
Paper crack	0.7393 $\pm$ 0.0067	0.7121 $\pm$ 0.0419	0.7240 $\pm$ 0.0146	0.6975 $\pm$ 0.0471
PVC pipe crack	0.9077 $\pm$ 0.0047	0.8834 $\pm$ 0.0127	0.8882 $\pm$ 0.0280	0.8345 $\pm$ 0.1029
Substrate gap [†]	20.2 pp	21.2 pp	18.1 pp	18.4 pp

[†]Dummy crack AP@50-95 minus Paper crack AP@50-95.

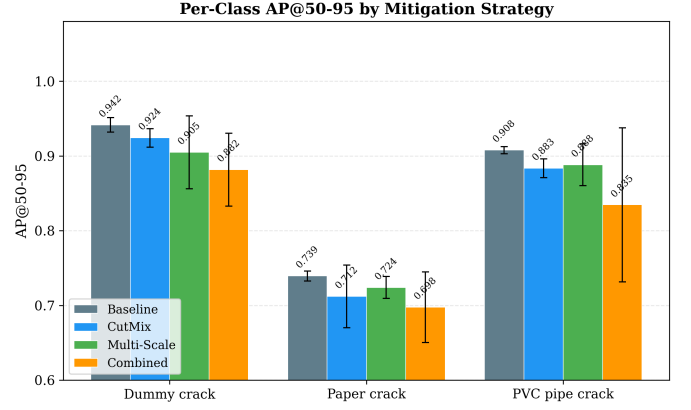


Fig. 5. Per-class AP@50-95 across mitigation strategies (mean $\pm$ std, 5 seeds). All three strategies degraded performance relative to baseline, with Paper crack consistently lowest.

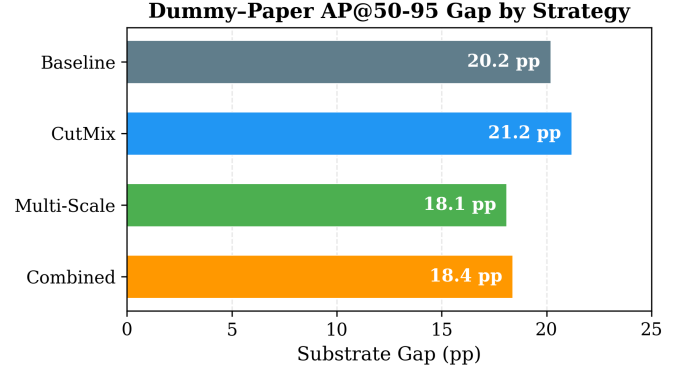


Fig. 6. Dummy–Paper substrate gap across mitigation strategies. Multi-Scale achieved the largest reduction (20.2 $\rightarrow$ 18.1 pp) but at the cost of degraded overall mAP.

V. DISCUSSION

A. Substrate Complexity as the Governing Factor

While textured backgrounds are known to degrade detection [4], [7], our contribution is quantifying this effect under controlled, class-balanced conditions—the 20–22 pp AP gap cannot be dismissed as a data imbalance artifact. Comparing AP@50 with AP@50-95 reveals the effect is primarily a *localization* problem (Fig. 7): paper crack maintains 97.1% AP@50 but drops to 74.0% at AP@50-95—a 23.1 pp gap—whereas dummy crack shows only 3.9 pp (seed 42; the same class hierarchy holds across all five seeds in Table II). The model detects paper cracks reliably but cannot draw tight bounding boxes where crack edges blend into the substrate. We note that annotation ambiguity may also contribute: bounding box placement on textured tissue is inherently less precise than on smooth PVC, so part of the AP@50-95 gap may reflect ground-truth noise rather than model error alone. Notably, dummy and PVC pipe cracks—both on smooth PVC—differ by only 3.4 pp despite distinct crack morphologies, whereas the substrate change from smooth PVC to textured tissue produces a 20+ pp gap. This confirms substrate texture, not

crack morphology, is the dominant factor. YOLO-World’s CLIP-based pre-training provides a measurable advantage (+0.83 pp mAP@50-95), but features transferred most effectively to high-contrast defects, indicating a fundamental visual domain challenge.

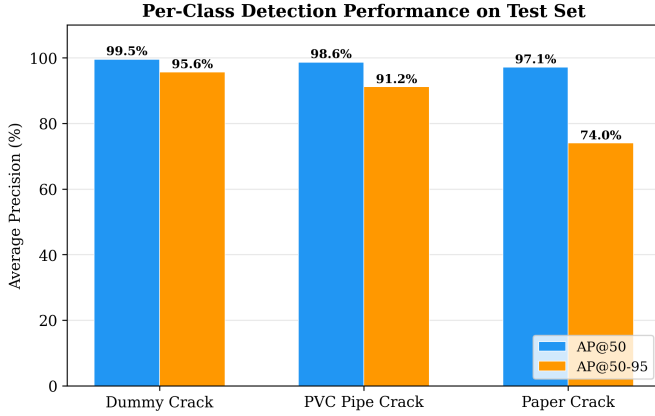


Fig. 7. Per-class AP at IoU thresholds 0.50 and 0.50–0.95 (YOLO-World XL, best single run, seed 42). Paper crack drops 23.1 pp vs. only 3.9 pp for dummy crack.

B. Limits of Augmentation-Based Mitigation

The failure of all three strategies to close the gap suggests the disparity is not primarily caused by insufficient scale or appearance diversity during training. Instead, the gap likely reflects a more fundamental challenge: paper crack annotations exhibit inherently lower contrast and less distinctive boundary features, making precise localization difficult regardless of augmentation.

This interpretation is supported by two observations. First, the localization gap (AP@50 minus AP@50-95) for Paper crack is substantially larger than for other classes across all strategies (23.1 pp vs. 3.9 pp for Dummy), confirming the detector finds Paper crack objects but cannot localize them precisely. Second, the Combined strategy degraded *all* classes uniformly rather than selectively improving Paper crack, with PVC pipe crack suffering the largest drop (−7.3 pp), suggesting aggressive augmentation harms fine-grained localization across the board. We note that Multi-Scale and Combined strategies used batch size 4 (vs. 8 for Baseline) due to memory constraints at higher resolution; noisier gradient estimates may partially explain their degraded performance, though CutMix (also batch 8) still underperformed the Baseline. Addressing the gap may therefore require strategies beyond training-time augmentation, such as substrate-specific fine-tuning, domain adaptation, or improved annotation protocols for low-contrast substrates.

C. Practical Implications

These findings have direct implications for deploying crack detection systems on real piping infrastructure. First, practitioners should expect significant performance variation across pipe materials and should evaluate detection models on each target substrate separately rather than relying on aggregate

mAP. Second, the localization nature of the gap means that detection-level alarms (at IoU 0.50) remain reliable even on challenging substrates, but applications requiring precise crack measurement or sizing should account for reduced bounding box accuracy on textured surfaces. Third, the failure of standard augmentation to close the gap suggests that material-specific training data or domain adaptation strategies may be necessary when deploying across heterogeneous pipe inventories.

D. Limitations

Several limitations should be considered when interpreting these results:

Test set size. The test set contains 108 images (~36 per class). While the multi-seed design controls for training stochasticity, per-class AP estimates carry sampling uncertainty that larger test sets would reduce. Confidence intervals around per-class AP could not be computed without bootstrapping over test images, which we leave to future work.

Controlled proxies vs. field specimens. The use of controlled proxies prioritizes internal validity (isolating substrate effects) over ecological validity (field realism). The specific magnitude of the AP gap may differ on field specimens where confounds—crack age, weathering, surface contamination, and non-uniform lighting—interact with substrate properties.

Confounds. Crack generation methods differ across classes (drawn lines vs. physical fractures vs. torn tissue), partially confounding texture with morphology. The 3.4 pp dummy–PVC gap (Section V-A) provides an upper bound on this morphology confound, which is small relative to the 20+ pp substrate gap. Additionally, all specimens share a blue color, limiting generalization to other pipe colors.

Model scope. Only two YOLO-family architectures were compared. The localization gap may manifest differently in two-stage detectors or transformer-based architectures that use different bounding box regression mechanisms.

VI. CONCLUSION AND FUTURE WORK

This paper investigated how substrate visual complexity affects crack detection under controlled data conditions. YOLO-World XL achieved mAP@50-95 of 0.8628 ± 0.0044 over five seeds, yet per-class AP ranged from 94.2% (dummy crack on smooth PVC) to 73.9% (paper crack on textured tissue) despite 400 labels per class. An IoU-threshold decomposition confirmed this is primarily a localization problem: paper crack retains 97.1% AP@50 but drops 23.1 pp at stricter thresholds, indicating the model detects cracks but cannot draw tight bounding boxes on textured substrates. A parallel comparison with YOLOv8x confirmed a consistent 20–22 pp gap across architectures. Three augmentation strategies failed to reduce the gap (Bonferroni-corrected $p > 0.05$), with the Combined strategy degrading mAP by 5.8 pp. These results demonstrate that substrate visual properties, rather than data quantity or model choice, govern detection difficulty—a fundamental localization challenge tied to low-contrast crack boundaries.

Future work should pursue several directions. First, field validation on naturally occurring pipe failures across diverse

materials (concrete, cast iron, clay) would establish ecological validity and calibrate the magnitude of the substrate gap under real-world conditions. Second, architecturally distinct detectors—including two-stage models (e.g., Faster R-CNN), transformer-based architectures (e.g., DETR), and segmentation models—would clarify whether the localization gap is specific to single-stage detectors or a more general phenomenon. Third, experiments that vary substrate texture while holding crack formation constant (e.g., identical cracks applied to smooth vs. rough surfaces) would fully disentangle texture from morphology effects. Fourth, targeted mitigation strategies such as domain adaptation, substrate-conditioned feature normalization, or pixel-level segmentation losses may prove more effective than the augmentation-based approaches tested here. Finally, the best-performing model can be integrated into a mobile robotic platform for autonomous pipe inspection [19]–[21], with substrate-aware confidence calibration enabling reliable deployment across diverse pipe inventories.

Data and Code Availability

The dataset is available on Roboflow Universe (<https://universe.roboflow.com/gazzard/pipe-crack-detection/dataset/1>) under CC BY 4.0. Code will be released at <https://github.com/gazzard-ust/crack-detection-analysis> upon publication.

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